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Exploring all-quantum long short-term memory networks to enhance text sentiment classification

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Abstract
Quantum natural language processing (QNLP) can efficiently represent and process in parallel a vast number of words and sentences in high-dimensional Hilbert space by leveraging the characteristics of quantum superposition and entanglement, thereby accelerating complex text sentiment classification tasks. This study, for the first time, proposes a full quantum model based on long short-term memory network for text sentiment classification, where the fundamental representation learning and sequence modeling operations are all realized through quantum state evolution. Additionally, a subspace projective quantum self-attention mechanism is designed to effectively capture long-distance dependencies and semantic associations in text sequences. This method achieves higher accuracy, reduced circuit complexity, and expanded scalability with larger datasets. Extensive experiments demonstrate that the proposed approach outperforms existing quantum-enhanced models in terms of overall performance and exhibits strong robustness against various quantum noise interferences. Our model is anticipated to be effectively implemented on noisy intermediate-scale quantum (NISQ) devices to yield results of practical significance, thereby advancing the development of the QNLP field.

Keywords Quantum natural language processing · Text sentiment classification · Quantum long short-term memory · Parameterized quantum circuits · Quantum image processing

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1 Introduction

In the field of natural language processing (NLP), text sentiment classification is a fundamental and challenging task, aiming to identify and extract emotional tendencies from text [1]. With the development of deep learning and the widespread adoption of attention mechanisms, convolutional neural networks (CNNs) and recurrent neural networks (RNNs) have significantly improved performance in text classification [2], sentiment analysis [3], and other related tasks [4]. Nevertheless, as the volume of data has surged exponentially and the complexity of texts has escalated, these methods have gradually revealed their limitations in handling long-distance dependencies and discerning complex patterns [5]. Furthermore, they entail substantial consumption of computational resources during the feature extraction stage.

Quantum computing, as a non-traditional technology, has demonstrated remarkable technical advantages and broad development prospects in the field of machine learning [6]. By virtue of properties such as quantum superposition and entanglement, it provides innovative solutions to challenges in traditional machine learning, such as high-dimensional data processing and complex feature extraction. As a crucial branch of machine learning, NLP is gradually becoming a key direction for the application of quantum computing technology. The representation capability of quantum states in high-dimensional Hilbert space enables the mapping of classical lexical semantics and sentence structures to quantum systems, and achieves efficient modeling of semantic relationships [7]. Current research mainly follows two strategies: one is to use quantum operations for text data preprocessing to build a hybrid quantum-classical classification model, and the other is to construct a complete quantum system to implement classification operations through the evolution of quantum states compiled from text data via quantum circuits.

In hybrid classification models, quantum-mechanical principles are incorporated into classical architectures to improve the efficiency of text sentiment classification tasks [8, 9]. One notable instance is from Shi et al. in 2021, where they proposed two end-to-end quantum heuristic deep neural network models. These models represent words as complex vectors, effectively simulating the superposition characteristics of quantum states to enhance the semantic expression dimension in text sentiment classification tasks [10]. However, the model relies entirely on complex number operations to simulate quantum information processing, resulting in increased computational complexity. A few years later, Li et al. introduced a quantum-inspired deep learning framework (named QITSA) that uses complex vectors and density matrices to construct a mixed state of words and sentences. It combines the self-attention mechanism and 2D-CNN to extract deep features, but the construction complexity of the overall model is relatively high [11]. In 2024, Yu et al. designed a quantum RNN based on parameter non-sharing batch upload strategies, which outperforms the previous quantum RNNs in text sentiment classification of low-resource languages [12]. Nonetheless, the batch processing approach may lead to a decline in computational efficiency when facing complex sentence structures. In summary, the current quantum-classical hybrid methods generally suffer from problems such as high computational complexity and information loss caused by repeated data form transformations.

Conversely, recent research focuses on constructing complete quantum models using parameterized quantum circuits [13] that can run on noisy intermediate-scale quantum (NISQ) [14] devices, fully leveraging quantum advantages to address text sentiment classification problems. In 2023, Lorenz et al. conducted the first simple NLP experiment on the NISQ device [15], demonstrating the development potential of quantum computing in the field of text sentiment classification. In the same year, Zheng et al. constructed a quantum model based on parametric quantum circuits [16], aiming to overcome the limitations of quantum models such as insufficient flexibility in text classification tasks. However, none of these research efforts has performed adequate experiments to confirm that the model retains high accuracy despite the effects of quantum noise. In 2024, Li et al. integrated the self-attention mechanism with quantum neural networks for the first time, providing new ideas for the field of quantum NLP (QNLP) [17]. The constraints of previous quantum inner product attention in capturing long-distance dependencies were addressed by using the Gaussian projection quantum self-attention mechanism, and the stability of the model under low-level quantum noise was verified through sufficient experiments. Nevertheless, the accuracy of the model was verified only on small-scale datasets. Taken together, these studies indicate that the existing full quantum methods are still in the initial stage of development, and have certain deficiencies in terms of classification accuracy, feasibility for large-scale datasets, and stability under the influence of noise.

To address the limitations of existing QNLP models, in this study, we propose an all-quantum long short-term memory (AQLSTM) network. By "all-quantum" we mean that core representational, attentional, and sequential memory-update operations are all-quantum-native, whereas classical components are retained merely as interfaces to classical input and label spaces. This model features low-complexity quantum circuits to boost the interaction among queries, keys, and values in the self-attention mechanism. We substitute the gating mechanism in LSTM with quantum circuits to create a comprehensive quantum model. By executing these operations at the quantum layer, key semantic information and long-range dependencies in the input sequence are effectively captured. To evaluate the model's performance, we conducted numerical experiments on publicly available datasets of varying scales. The results demonstrate that this model offers superior classification performance, remarkable scalability, and notable robustness against various forms of quantum noise. Our primary contributions are as follows:

- We propose an all-quantum LSTM model for text sentiment classification, that performs key representation learning and sequence modeling entirely in the quantum domain, thereby improving the classification accuracy of existing quantum models.
- We develop a subspace projective quantum self-attention (SP-QSA) mechanism that utilizes quantum superposition and entanglement to capture complex dependency relationships in sequences, while minimizing circuit complexity.
- We enhance the QNLP model to accommodate larger-scale datasets, demonstrating that, under the influence of three classical forms of noise and their various combinations, the model displays impressive robustness.

The structure of this paper is organized as follows: Sect. 2 presents the foundational theories of quantum computing relevant to text sentiment classification. Section 3 elaborates on the proposed AQLSTM architecture, along with the design specifics of the quantum circuits. Section 4 outlines the experimental settings and numerical results from comparative experiments. Finally, Sect. 5 concludes the paper and explores potential future research directions.

2 Fundamentals of quantum computing in text sentiment classification

Prior to exploring the AQLSTM model, it is essential to first grasp fundamental concepts of quantum computing and their specific application methods in text sentiment classification.

2.1 Quantum states for semantic representation

The quantum state is the foundational element of quantum computing, allowing a qubit to exist in any superposition, which enables parallel computation within quantum systems. Specifically, by means of quantum entanglement, multiple qubits can interact to create a completely new state. The state of a quantum bit can be represented as a vector in a two-dimensional Hilbert space:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \tag{1}$$

where $|0\rangle$ and $|1\rangle$ are the calculation basis vectors; α and β represent the probability amplitudes of $|0\rangle$ and $|1\rangle$, respectively, and $|\alpha|^2 + |\beta|^2 = 1$. For n qubits, their states can be represented as a 2^n dimensional vector:

$$|\Psi\rangle = \sum_{i=0}^{2^n-1} c_i|i\rangle, \tag{2}$$

where c_i denotes the probability amplitude associated with the basis state, with $c_i \in C$ and $\sum_{i=0}^{2^n-1} |c_i|^2 = 1$. The state $|i\rangle$ corresponds to the calculated basis state of all combinations of n qubits. For instance, for two qubits, the basis state is $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$.

In the task of text sentiment classification, we encode the word vectors of the text into quantum states via quantum embedding. Not only are the semantic features of the words retained, but also the text information is encoded into the high-dimensional Hilbert space by using the quantum superposition property, achieving a more efficient representation method. In this way, multiple words in the text are mapped to different quantum states, laying the foundation for the subsequent simulation of semantic relationships between words in quantum circuits.

2.2 Quantum gates for simulating semantic relationships

A quantum gate is a linear, unitary operator that acts on a quantum state, and its effect can be expressed as the product of a matrix and a quantum state:

$$|\psi'\rangle = U|\psi\rangle, \tag{3}$$

where $U^{\dagger}U = I$. The quantum gates utilized in this study consist of the Hadamard gate (H), CNOT gate ($CNOT$), Pauli X gate (R_X), Pauli Y gate (R_Y), and Pauli Z gate (R_Z) [18], as illustrated in the implementation of subsequent quantum circuits.

For text information encoded in quantum states, we operate on it using quantum gates to simulate the mutual influence relationship between text vocabularies. By combining varying quantum gates and relying on the characteristics of quantum entanglement, the complex semantic relationships between words can be simulated in a high-dimensional space. The evolution mechanism driven by quantum gates enables the efficient propagation of text information among qubits and also allows for the capture of more refined context dependencies.

2.3 Quantum measurement for semantic feature extraction

Quantum measurement is used to extract information from quantum states. It is usually achieved by calculating the expected value of the Pauli operator. For example, measuring the expected value of a quantum bit is:

$$\langle Z\rangle = \langle\psi|Z|\psi\rangle, \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \tag{4}$$

where $\langle\psi|$ is the conjugate transpose of $|\psi\rangle$ and $\langle\psi| = |\psi\rangle^{\dagger}$. For multi-qubit systems, the measurement operator is the tensor product of each qubit operator: $Z_1 \otimes Z_2 \otimes \cdots \otimes Z_n$. Given a quantum state in the form:

$$|\Psi\rangle = \sum_{i=0}^{2^n-1} c_i|i\rangle, \tag{5}$$

the expected value of the measurement operator is then given by:

$$\langle Z\rangle = \sum_{i=0}^{2^n-1} |c_i|^2(-1)^{\text{bit-sum}(i)}, \tag{6}$$

where bit-sum(i) signifies the number of $|1\rangle$ in the binary representation.

After the quantum state of text information evolves via quantum gates, it must be converted into classical information for classification decisions. Key features of the quantum state are extracted by measuring specific quantum operators, and these measurement results reveal the semantic structure and emotional characteristics of the

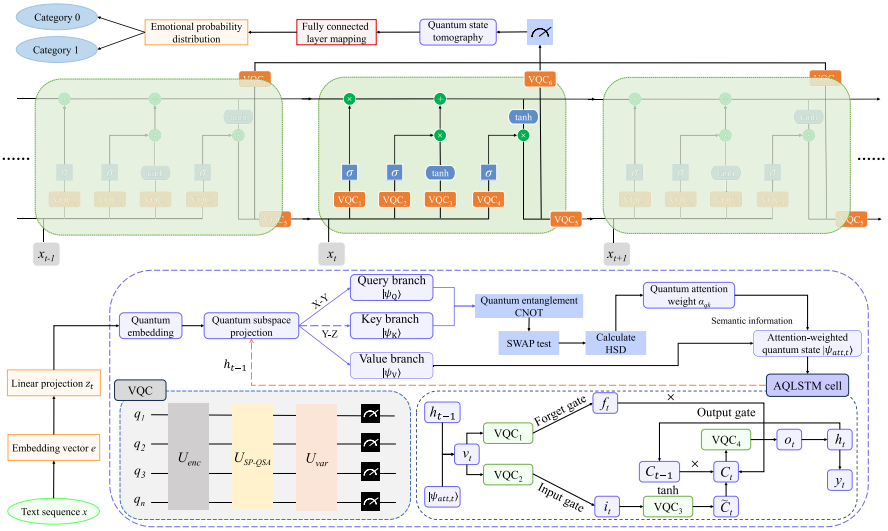


Fig. 1 Architecture of the AQLSTM model including quantum embedding layer, SP-QSA mechanism, and quantum gate memory unit

input text in its quantum representation, which are then used to compute classification probabilities. Through quantum measurement, the model maps high-dimensional quantum information to interpretable emotional labels, thus completing the final step of the text sentiment classification task.

3 Quantum architecture design and circuit implementation

In this section, we will provide a comprehensive introduction to the proposed AQLSTM network, including the specific design of the variational quantum circuit (VQC). This design encompasses rotational quantum embedding, a SP-QSA mechanism, and variable layering techniques.

3.1 Overall architecture of the AQLSTM model

The proposed AQLSTM model establishes a novel synergy between quantum-enhanced sequence modeling and self-attention-driven contextual awareness. As illustrated in Fig. 1, the framework comprises three core quantum modules: quantum embedding layer for high-dimensional semantic representation, quantum self-attention mechanism for contextual dependency modeling, and quantum gated memory unit for sequential information retention.

The input text sequence first undergoes quantum embedding via parameterized rotation gates (Sect. 3.2), projecting classical word vectors into a high-dimensional Hilbert space where semantic relationships manifest as quantum state superpositions. The quantum self-attention module (Sect. 3.3) then processes these states by way of quan-

tum circuits implementing Query-Key-Value operations, harnessing entanglement to compute attention weights via Hilbert-Schmidt distance measurements. This quantum attention layer dynamically modulates information flow to the subsequent Quantum LSTM unit (Sect. 3.4), which employs variational quantum circuits to replace classical neural network components in the gating mechanism. The architecture iteratively processes sequential inputs across time steps while maintaining quantum state coherence, with final measurements feeding into classical activation functions for prediction tasks.

Specifically, the term all-quantum refers to the fact that all core representation-learning and sequence-modeling operations, including the encoding of classical embeddings into Hilbert-space quantum states, self-attention computation, and gated memory updates, are implemented through parameterized quantum circuits acting on quantum states. The overall information flow follows a classical-to-quantum-to-classical paradigm: classical word vectors are first encoded into quantum states via rotational embedding; subsequent feature extraction, attention modulation, and temporal gating are then carried out within the quantum processing core through unitary evolution and entangling operations; finally, measurement-based readout transforms the terminal hidden state into a classical vector for the Softmax classifier.

This design is fundamentally distinct from quantum-inspired models, which simulate quantum behavior through classical algorithms without constructing explicit quantum circuits, and from hybrid quantum-classical models, which introduce quantum-enhanced modules into otherwise classical architectures. The remaining classical components in our pipeline, namely the preprocessing embedding lookup, the Adam optimizer used for parameter updates, and the final Softmax layer, serve only as necessary interfaces between the quantum processing core and the classical data and label spaces, rather than as constituents of the model's core representational and sequential inference capacity. We thus label the model "all-quantum" in that the entire learning pipeline from feature representation to sequence modeling is quantum-native in its core operations.

3.2 Rotational quantum embedding

In NLP tasks, the efficiency of word embedding directly affects the performance of subsequent models. Classical embedding methods capture semantic relationships between words by mapping them to fixed vector spaces. However, the expressive power of classical embedding is limited by the vector dimension and cannot fully express complex sentence structures. We therefore employ the quantum embedding method to encode classical physical data into quantum states via parameterized rotation gates, thereby laying the groundwork for subsequent quantum operations.

In particular, for the input text sequence $\mathbf{x} = [x_1, x_2, \ldots, x_T]$ that has been mapped to the embedding vector $\mathbf{e} = [e_1, e_2, \ldots, e_T]$ after classical embedding, where $e_t \in \mathbb{R}^{d_{\text{embedding}}}$, these embedding vectors are then transformed linearly to a form suitable for quantum embedding:

$$z_t = W_e e_t + b_e, \tag{7}$$

where $W_e \in \mathbb{R}^{2n \times d}$ is the embedding projection matrix and $b_e \in \mathbb{R}^{2n}$ is the bias term. Here, the dimension $2n$ directly corresponds to the number of qubits n: each qubit requires two rotation angles (e.g., R_Y and R_Z) to fully parameterize its quantum state on the Bloch sphere. This design ensures that the d-dimensional classical vector e_t is projected into a $2n$-dimensional space, where every dimension controls a specific rotational degree of freedom of the quantum system.

Due to the wide adaptability and extensibility of angle embedding, we embed each dimension of the classical vector z_t into the rotation angle of the quantum state, forming the quantum state $|\psi_t\rangle$. A parameterized revolving gate of R_Y and R_Z is used on each qubit to achieve this conversion:

$$|\psi_t\rangle = \bigotimes_{i=1}^{n_{\text{qubits}}} R_Y(z_{t,2i-1}) R_Z(z_{t,2i}). \tag{8}$$

This quantum embedding operation is more than just a formal data transformation; it essentially allows the model to achieve a richer representation of textual emotional features in high-dimensional Hilbert space. Compared with traditional word embedding methods, quantum embedding can encode nonlinear semantic relationships between words through quantum superposition and entanglement, which has significant advantages in capturing complex emotional semantics such as sarcasm and rhetorical questions. Moreover, the higher dimensionality of representation enhances the model's generalization ability when dealing with diverse texts, such as movie reviews, social commentaries, and user feedback. This, in turn, improves the overall accuracy of emotional classification.

3.3 Subspace projective quantum self-attention mechanism

Although recent studies have partially implemented the classical self-attention mechanism-specifically the Q-K-V branches and attention score computation-within quantum circuits, they typically treat the Q, K, and V projections as independent subcircuits. After measurement, the outputs are returned to the classical layer and normalized via a Softmax function, lacking a physical mechanism to capture long-range semantic dependencies between Q and K. In contrast, the SP-QSA mechanism proposed in this study introduces repeatable CNOT-based entanglement units between the Q and K branches at the quantum layer, explicitly encoding cross-token dependencies. This fundamentally enhances the model's ability to recognize complex emotional constructs such as negation and contrast. Furthermore, the attention module computes all Q-K similarity scores using a single SWAP-test measurement, thereby avoiding frequent quantum-classical data interactions.

In classical vector spaces, sentiment features typically manifest as specific dimensional combinations of vectors. Under quantum representation, these characteristic combinations are mapped to phase and amplitude configurations of quantum states. Through attention-weight modulation, the amplitude of each basis state in quantum superposition is dynamically amplified or attenuated, forming a "sentiment-focused" substate. The corresponding subspace of this substate constitutes the "semantic sub-

space," indicating that during quantum measurement or subsequent gating operations, components within this subspace are preferentially activated to amplify sentiment-specific features. This significantly enhances both discriminative efficiency and classification accuracy in sentiment analysis tasks.

In detail, we employ distinct combinations of quantum rotation gates to generate differentiated representations for Q, K, and V. The $R_X + R_Y$ combination for Q induces a superposition state along the X-Y plane of the Bloch sphere, while $R_Y + R_Z$ for K creates a bias toward the Y-Z plane. This specialized projection strategy maps queries and keys (Q/K) into distinct subspaces within the Hilbert space, thereby enhancing the model's ability to discriminate semantic differences-particularly in texts with ambiguous affective expressions or high degrees of semantic polysemy, such as ironic constructions exemplified by utterances like "This movie is so good it's nauseating." For each time step t, the embedded quantum state $|\psi_t\rangle$ is passed separately by the mathematical formulations below.

Query (Q): Transform through R_X and R_Y gates,

$$|\psi_Q\rangle = |\psi_t\rangle \bigotimes_{i=1}^{n_{\text{qubits}}} R_X(w_{Q,i}) R_Y(w_{Q,i}). \tag{9}$$

Key (K): Transform through R_Y and R_Z gates,

$$|\psi_K\rangle = |\psi_t\rangle \bigotimes_{i=1}^{n_{\text{qubits}}} R_Y(w_{K,i}) R_Z(w_{K,i}). \tag{10}$$

Value (V): Transform through R_Z and R_X gates,

$$|\psi_V\rangle = |\psi_t\rangle \bigotimes_{i=1}^{n_{\text{qubits}}} R_Z(w_{V,i}) R_X(w_{V,i}). \tag{11}$$

Here, $w_{Q,i}, w_{K,i}, w_{V,i} \in \mathbb{R}$ is a trainable rotation angle parameter. The quantum state characterization of Q/K/V is dynamically adjusted by gradient optimization.

In text sentiment classification tasks, sentiment features are seldom determined by adjacent lexical items but rather depend on long-range dependency relationships between words positioned distantly within sentences. By applying CNOT gates between adjacent qubits to establish inter-qubit entanglement correlations (as shown in Fig. 2), the query state $|\psi_Q\rangle$ and key state $|\psi_K\rangle$ are enabled to effectively encode long-range semantic dependencies via quantum entanglement, thereby enhancing context-sensitive sentiment discernment capabilities for complex linguistic contexts.

In the classical attention mechanism, the core operation is to determine the attention weight by calculating the similarity between the query vector Q and the key vector K. In order to realize the quantization extension, we design a physically achievable quantum similarity measurement scheme based on Hilbert-Schmidt distance (HSD) [19], which is mathematically equivalent to the classical dot-product attention. The

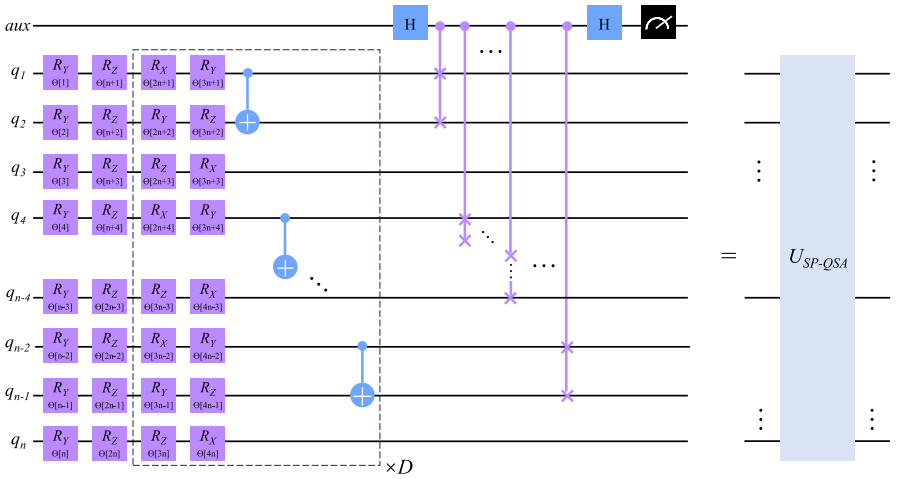


Fig. 2 Quantum circuit structure of SP-QSA mechanism. The dashed box operation can be executed D times to enhance word feature extraction

HSD is a measure between two quantum states (the density matrix), defined as:

$$D_{\mathrm{HS}}(\rho, \sigma) = \sqrt{\mathrm{Tr}[(\rho - \sigma)^2]}, \tag{12}$$

where ρ and σ are two density matrices, and Tr represents trace operations. For the pure state $|\psi_q\rangle\langle\psi_q|$ and $|\psi_k\rangle\langle\psi_k|$, the above equation can be simplified as:

$$D_{\mathrm{HS}} = \sqrt{2(1 - |\langle\psi_q|\psi_k\rangle|^2)}, \tag{13}$$

which provides a theoretical basis for indirect measurement of HSD by quantum SWAP test. Furthermore, HSD enables the detection of global state disparities, extracting deeper semantic information at lower computational cost compared to traditional dot-product attention mechanisms. This capability empowers the model to more accurately identify sentiment polarity when processing syntactically complex constructs containing contextual shifts, negations, and metaphorical expressions, thus enhancing both classification performance and system robustness.

To preserve quantum state integrity while evaluating state similarity, we adopt the quantum SWAP test protocol with ancilla-assisted measurement.

Experimental measurement: The composite system is constructed using two quantum states $|\psi_q\rangle$ and $|\psi_k\rangle$:

$$|\psi\rangle = |0\rangle_a \otimes |\psi_q\rangle \otimes |\psi_k\rangle, \tag{14}$$

where $|0\rangle_a$ indicates the auxiliary qubit. The SWAP test circuit performs the following operations:

1) A Hadamard gate is applied to the auxiliary qubit:

$$H|0\rangle_a = \frac{|0\rangle + |1\rangle}{\sqrt{2}}. \tag{15}$$

2) A controlled-SWAP operation is employed to swap $|\psi_q\rangle$ and $|\psi_k\rangle$ conditioned on auxiliary state $|1\rangle$.
3) The Hadamard gate is again applied to the auxiliary qubit, and the final state is:

$$\begin{aligned} |\psi_f\rangle = \frac{1}{2}\big[&|0\rangle_a(|\psi_q\psi_k\rangle + |\psi_k\psi_q\rangle) \\ &+ |1\rangle_a(|\psi_q\psi_k\rangle - |\psi_k\psi_q\rangle)\big]. \end{aligned} \tag{16}$$

HSD Calculation: The expected value of Pauli-Z for the auxiliary qubit is measured as:

$$\langle Z_a\rangle = |\langle\psi_q|\psi_k\rangle|^2. \tag{17}$$

This gives the HSD measurement result:

$$D_{HS} = \sqrt{2(1 - \langle Z_a\rangle)}. \tag{18}$$

To establish compatibility with the classical attention mechanism, we show that there is a monotonic relationship between HSD and classical cosine similarity when $|\psi_q\rangle$ and $|\psi_k\rangle$ are normalized quantum states (see Appendix A). Based on this, the attention weight is defined as:

$$\alpha_{qk} = \frac{\exp(-\gamma D_{\mathrm{HS}}(\rho_q, \rho_k))}{\sum_k \exp(-\gamma D_{\mathrm{HS}}(\rho_q, \rho_k))}, \tag{19}$$

where γ is a temperature parameter used to adjust the steepness of the attention distribution, and the negative exponential function $\exp(-D_{\mathrm{HS}})$ maps the HSD as a probability distribution. The design realizes the continuous mapping of quantum state similarity to probability distribution by parametrizing HSD, and guarantees the normalization condition of $\sum_k \alpha_{qk} = 1$.

The resulting attention weights are then applied to the value state, which indirectly affects subsequent status updates in AQLSTM. This design allows the model to dynamically adjust its attention to the input sequence at each time step, boosting the model's ability to capture long-distance dependencies.

3.4 Implementation of quantum gated memory unit

The proposed variational quantum circuit architecture comprises four coherent functional modules, namely quantum embedding layer, quantum self-attention mechanism layer, variable layer (dotted line box), and quantum measurement layer. By integrating SP-QSA into AQLSTM and closely integrates it with the quantum gating mechanism,

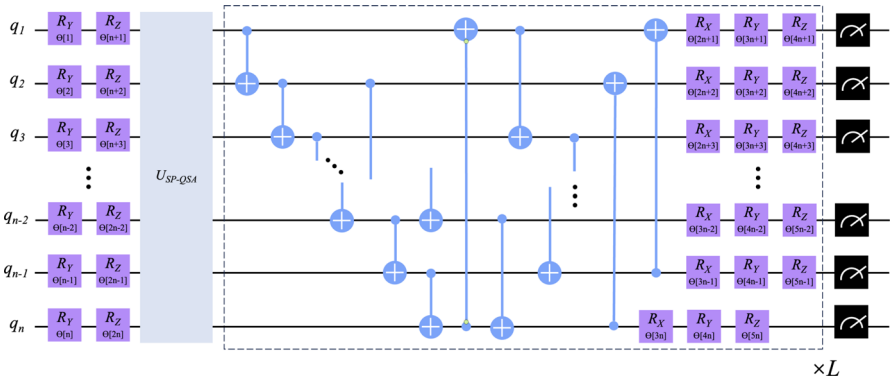


Fig. 3 Quantum circuit structure of the AQLSTM model. The dashed box operation can be executed L times to enhance the modeling of sentence sequences

SP-QSA directly acts on the input at each time step t. This enhances information interaction between different time steps and captures complex dependencies in the input sequence using quantum entanglement. The overall structure is shown in Fig. 3. Increasing the number of repeat layers L can enhance the depth of quantum circuits, and improve the modeling ability of complex timing patterns via multiple parametric unitary transformations.

Since the gating mechanism of classical LSTM is essentially a cascade of nonlinear transformations, we construct its quantum correspondence by parameterizing the unitary transformation properties of quantum circuits. A recursive architecture with quantum state evolution is constructed by connecting multiple VQC modules via temporal topology, in which the entanglement gate design of each module ensures the quantum correlation between hidden states. The parametric quantum circuit forms a nonlinear mapping equivalent to the classical activation function within the desired measurement range. This is achieved through the nonlinear state evolution generated by the controlled revolving gate and entanglement operations. As illustrated in Fig. 4, the yellow dashed box indicates the cell state, the red highlights the forgetting gate, the blue denotes the input gate, and the green represents the output gate.

To further enhance the capabilities of long-sequence modeling, we reformulate the gating mechanisms and derive the associated formulas, highlighting the specific role of attention-weighted quantum semantic information in text sentiment classification tasks. For the output attention weight $\alpha_{t,k}$, corresponding to value state $|\psi_{V_k}\rangle$, its attention-weighted quantum state is defined as:

$$|\psi_{\text{att},t}\rangle = \sum_{k=1}^{K} \alpha_{t,k} |\psi_{V_k}\rangle . \tag{20}$$

In sentiment classification tasks, this attention-weighted quantum state selectively filters sentiment-relevant semantic subspaces associated with the current lexical unit while mitigating information redundancy. We leverage $|\psi_{\text{att},t}\rangle$ semantic information to replace conventional embedding states $|\psi_t\rangle$ and combine it with the previous hidden

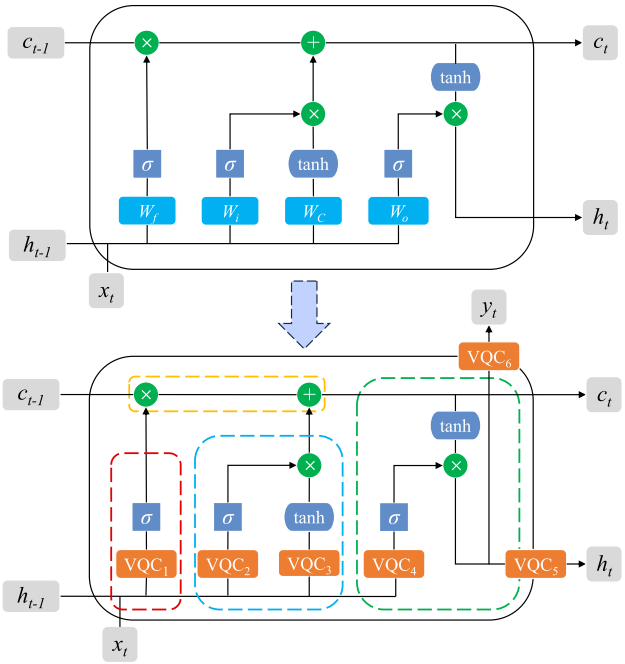


Fig. 4 Evolution of LSTM units: From classical to quantum

state h_{t-1} via tensor product to form the gated input:

$$v_t = \left|\psi_{\text{att},t}\right\rangle \otimes |h_{t-1}\rangle\,, \tag{21}$$

where $|h_{t-1}\rangle$ denotes the vectorized quantum hidden state at the previous time step, and $\otimes$ represents the tensor product operation. This step ensures the gating signal can synchronously acquire information from both quantum attention-weighted semantic states and historical sentiment memory.

Four gating sub-circuits $U_f(\theta_f)$, $U_i(\theta_i)$, $U_C(\theta_C)$, and $U_o(\theta_o)$ are created using trainable parameterized quantum circuits, where $\theta_\triangle$ denotes the set of trainable rotation angle parameters for each corresponding circuit. The Pauli-Z expectation value is measured on the input state v_t for each sub-circuit to obtain the gating signals:

$$\begin{aligned} f_t &= \sigma\left(\langle Z\rangle_{U_f(\theta_f)}(v_t)\right), \\ i_t &= \sigma\left(\langle Z\rangle_{U_i(\theta_i)}(v_t)\right), \\ o_t &= \sigma\left(\langle Z\rangle_{U_o(\theta_o)}(v_t)\right), \\ C_t &= \tanh\left(\langle Z\rangle_{U_C(\theta_C)}(v_t)\right), \end{aligned} \tag{22}$$

where $\langle Z\rangle_{U_\triangle}(v_t) = \langle v_t|U_\triangle^\dagger Z U_\triangle|v_t\rangle$ and $\tanh(x) = (e^x - e^{-x})/(e^x + e^{-x})$. Additionally, $\sigma(x) = 1/(1+e^{-x})$ is the Sigmoid activation function, which guarantees the gating values are constrained within the interval (0, 1).

Algorithm 1: Training algorithm of the AQLSTM network

Input: Number of qubits n_q, number of quantum layers n_l, input sequence $\mathbf{x} = [x_1, x_2, \ldots, x_T]$, label y, epoch, batch size B

Output: Trained AQLSTM model parameters

Initialize ansatz parameters θ and weights w from a Gaussian distribution $\mathcal{N}(0, 0.01)$, and set bias b to 0 **for** *epoch* $= 1, 2, \ldots, E$ **do**

  Prepare the training data batch B **for** *each sample* $(\mathbf{x}, y)$ *in* B **do**

    Initialize the hidden state $h_t \leftarrow \mathbf{0}$ and cell state $c_t \leftarrow \mathbf{0}$ **for** $t = 1$ *to* T **do**

      Encode each input x_t into a quantum state $|\psi_t\rangle$ via Eq. (8) **for** *Quantum layer* $D = 1$ *to* D **do**

        Generate attention components $|\psi_Q\rangle, |\psi_K\rangle, |\psi_V\rangle$ via Eqs. (9)-(11) Establish Q-K entanglement by using CNOT gate

      Perform the SWAP test via Eqs. (14)-(17) Calculate the Hilbert-Schmidt distance D_{HS} via Eq. (18) Obtain attention weight by using HSD: $\alpha_t \leftarrow \text{Softmax}(-\gamma D_{HS})$ Evaluate the VQC gating signal via Eq. (21) Renew cell state c_t via Eq. (23) Generate output gate: $o_t \leftarrow \sigma(\langle Z \rangle U_o(\theta_o)|\psi_V\rangle \otimes h_{t-1})$ Update hidden status: $h_t \leftarrow o_t \odot \tanh(c_t)$

    Compute $\mathcal{L} \leftarrow \text{CrossEntropy}(y, h_T)$; `// Cross-entropy loss`

    Update θ using optimizer $\mathcal{O}$

The forget gate determines which emotional information is discarded, enabling the model to focus more effectively on the core emotional cues that genuinely influence subsequent classification decisions. The input gate controls which new emotional features at the current step $\left|\psi_{\text{att},t}\right\rangle$ are incorporated into the cell state c_t, preventing the model from introducing all new information at once and thereby avoiding overfitting or conflicts in emotional polarity. The candidate cell state is generated by applying a nonlinear transformation to the current attention-focused quantum semantic state and the historical emotional state. The output gate regulates how much emotional information is released from the candidate cell state c_t to the hidden state h_t, which is ultimately passed to the classifier. In long sentences, this mechanism facilitates dynamic modification of outputs at different positions, helping the classifier to capture key information more effectively.

Following the computation of the forget gate f_t, input gate i_t, candidate state C_t, and output gate o_t, the state update operation is performed:

$$\begin{aligned} c_t &= f_t \cdot c_{t-1} + i_t \cdot \widetilde{C}_t, \\ h_t &= o_t \cdot \tanh\left(c_t\right), \end{aligned} \tag{23}$$

where c_{t-1} denotes the cell state from the preceding timestep; h_t and c_t represent the quantum-enhanced hidden state and updated cell state at current time step t, respectively. The terminal quantum hidden state h_T at the end of the sequence serves as the global sentiment feature vector, which is fed into the parameterized classifier:

$$y = \text{Softmax}\left(W_c h_T + b_c\right), \tag{24}$$

where $W_c \in \mathbb{R}^{m \times d}$ and $b_c \in \mathbb{R}^m$ represent the classifier's weight matrix and bias vector, respectively, with the output dimension m corresponding to the number of sen-

Table 1 Quantum resource scaling of AQLSTM with sequence length T and qubit count n

Resource	Scaling	Experimental Value
Qubits	$n+1$ (independent of T)	5
Two-qubit gates per step	$O((D+4L+1)\cdot n)$	$\sim$24
Total circuit executions	$O(T)$	T

timent categories. Additionally, $h_T \in \mathbb{R}^d$ signifies the classical vector representation obtained through quantum state tomography of the hidden quantum state.

Each gating subcircuit operates on the attention-weighted input state v_t. In contrast to traditional QLSTM models that only process the original embedding, our model tightly couples the gating signal with the attention weights. This design allows each gate to open and close in a manner that more accurately reflects semantic focus and historical emotional context, thereby enhancing the sensitivity and adaptability of emotional feature control in the classification task.

The complete AQLSTM training algorithm is presented in Algorithm 1, which realizes quantum-enhanced modeling of text sequences via the collaborative work of the quantum embedding layer, SP-QSA mechanism, and quantum gated memory unit. At each time step, the SP-QSA module executes $O(Dn)$ two-qubit gate operations, the four gating sub-circuits contribute $O(4Ln)$ two-qubit gate operations, and the SWAP test adds $O(n)$ two-qubit gate operations, yielding a per-step cost of $O((D+4L+1)\cdot n)$. Since the model processes B samples each of length T per epoch, the overall per-epoch computational complexity is given by

$$O\big(B \cdot T \cdot (D+4L+1) \cdot n\big). \tag{25}$$

This result characterizes the computational cost of the proposed model across one training epoch.

3.5 Quantum resource scaling analysis

To facilitate assessment of near-term hardware feasibility, we briefly summarize how the key quantum resources of AQLSTM scale with sequence length T and qubit count n, as shown in Table 1. The AQLSTM architecture reuses the same set of $n+1$ qubits (including one ancilla for the SWAP test) across all time steps, so the qubit count remains fixed regardless of the input sequence length. The two-qubit gate count per time step is determined by the number of SP-QSA repetition layers D, VQC repetition layers L, and qubit count n, as detailed in the computational complexity analysis of Algorithm 1, notably in Eq. (25). Since the LSTM processes tokens sequentially, the total number of circuit executions grows linearly with T. For our experimental configurations ($n=4$, $D=L=1$), the model requires only 5 qubits and approximately 24 two-qubit gates per time step, indicating that the quantum resource requirements remain modest and compatible with the scale of current NISQ devices.

Table 2 Dataset sizes and partitioning details for the large-scale datasets

Dataset	Train	Validation	Test	Total
MPQA	8496	1035	1072	10603
SUBJ	8000	1000	1000	10000

4 Experimental results and performance analysis

In this section, we compare AQLSTM with existing related models on multiple public datasets of different scales to evaluate its performance and applicability. We then assess the robustness of the proposed model under quantum noise through dedicated noise experiments. Finally, through ablation studies, we demonstrate the effectiveness of the proposed SP-QSA mechanism in boosting overall model performance. All experiments are implemented using the PyTorch and PennyLane frameworks.

4.1 Experimental background

To comprehensively evaluate the model's performance, we conducted experiments using small-, medium-, and large-scale text sentiment classification datasets for comparative analysis. The small-scale datasets meaning classification (MC) and relative pronoun (RP) were sourced from [20]. The MC dataset contains 130 sentences constructed from 17 distinct words, with each sample containing 3-4 words. It was divided into 70 training samples, 30 validation samples, and 30 test samples. The RP dataset consists of 105 sentences formed from 115 words, where each sample contains 4 words, and was divided into 74 training samples and 31 test samples.

The Yelp, IMDb, and Amazon datasets are adopted as medium-scale datasets [21]. Each dataset involves 1,000 samples with a ratio of positive (1) to negative (0) labels (1:1), where each sample comprises a varying number of words ranging from several to dozens. For these three datasets, we randomly split 80% of the samples as training set and 20% as testing set.

For large-scale datasets, we selected opinion polarity dataset (MPQA) [20] and subjectivity dataset (SUBJ) [22]. The specific dataset sizes and divisions are presented in Table 2. It is worth noting that in traditional text sentiment classification tasks, 10,000 samples are usually not regarded as large-scale. However, the existing research in the field of quantum text sentiment classification has not yet been well experimentally verified on datasets of this scale.

For the small- and medium-scale datasets, we conducted nine independent runs with different parameter initializations. The mean accuracy was used as the final evaluation metric, and the standard deviation was reported to assess the stability of the experimental results. For the large-scale datasets, both accuracy and F1-score were adopted as evaluation metrics. The Adam optimizer was used for parameter optimization, and the cross-entropy loss function was employed throughout the experiments:

$$\mathcal{L} = -\frac{1}{B}\sum_{i=1}^{B}\sum_{c=1}^{C} y_{i,c}\log(p_{i,c}), \tag{26}$$

Table 3 Hyperparameter configuration in various datasets

Dataset	n	D	L	Learning rate
MC	2	1	1	0.001
RP	4	1	1	0.001
Yelp	4	1	1	0.0005
IMDb	4	1	1	0.0005
Amazon	4	1	1	0.0005
MPQA	4	2	1	0.001
SUBJ	4	2	1	0.001

where B is the batch size, C is the number of classes, $y_{i,c}$ is the true label of the i sample, and $p_{i,c}$ is the prediction probability of the i sample for class c. In quantum models, the number of qubits, circuit depth, and learning rate are critical hyperparameters, and their specific configurations are summarized in Table 3, where D and L represent the number of circuit layers, respectively.

4.2 Results and comparisons

We conducted comparative experiments on the small-scale MC and RP datasets using DisCoCat [15], QSANN [17], QSAMo [23], and QSAMAN [24] as baseline models. Since the number of two-qubit gates directly affects circuit complexity, evaluating quantum models in terms of both qubit count and two-qubit gate count provides a more informative measure than relying solely on the number of trainable parameters.

As shown in Table 4, our model achieves 100% test accuracy on the MC dataset while using fewer quantum gates than the competing methods. On the RP dataset, it delivers the best performance on both the training and test sets, reaching 100% and 80.65% accuracy, respectively. The substantial gap between training and test performance on RP is mainly attributable to the discrepancy in vocabulary distribution between the two splits, where most words appear only in the test set but not in the training set. In addition, the model converges rapidly on the small-scale datasets, as illustrated by the loss curves in Fig. 5, which decay quickly within the first 50 epochs and then stabilize. Overall, these results demonstrate the strong text sentiment classification capability of AQLSTM on small-scale datasets.

For the medium-scale datasets, we performed a detailed comparison with CSANN and QSANN. The results in Table 5 show that, compared with traditional methods, AQLSTM reduces the number of trainable parameters by an order of magnitude, requires 20% fewer quantum gates than existing quantum models, and achieves higher test accuracy. On the IMDb and Amazon datasets, the model also exhibits smaller standard deviations (1.33% and 0.88%, respectively), indicating greater stability of the experimental results. Benefiting from the strong feature-extraction capability of the SP-QSA mechanism, AQLSTM achieves higher classification accuracy with fewer parameters and better stability on medium-scale datasets.

At present, research on large-scale datasets in QNLP remains limited. We therefore conducted comparative experiments on the large-scale MPQA and SUBJ datasets using

Table 4 Numerical experiments were performed on MC and RP datasets

Method	MC			RP			
	#Paras	Complexity*	Test Accuracy(%)	#Paras	Complexity*	Train Accuracy(%)	Test Accuracy(%)
DisCoCat	40	/	79.80	168	/	90.60	72.30
QSANN	25	24/24	100.0	109	48/240	95.35 ± 1.95	67.74 ± 0.00
QSAMo	26	24/24	100.0	32	48/48	/	74.19 ± 0.00
QSMAN	18	/	100.0	137	/	97.30 ± 0.00	77.42 ± 0.00
QSALSTM	39	**19/19**	**100.0**	78	**39/39**	$\mathbf{100.0 \pm 0.00}$	$\mathbf{80.65 \pm 0.00}$

* The number of qubits and two-qubit quantum gates. The best experimental results are shown in bold

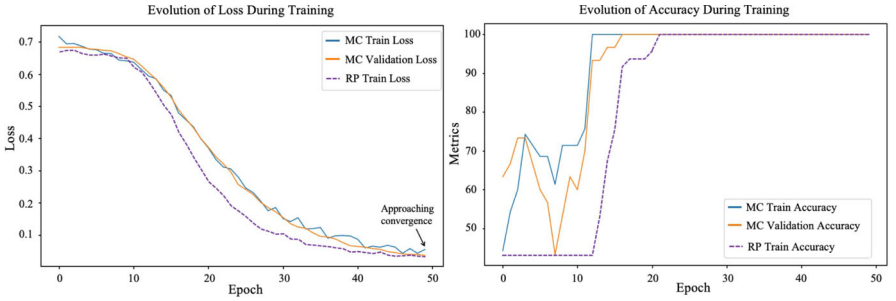


Fig. 5 Evolution of training/validation loss and accuracy metrics for both MC and RP datasets

Table 5 Numerical experiments performed on Yelp, IMDb, and Amazon datasets

Method	Yelp			IMDb			Amazon		
	#Paras	Complexity*	Accuracy(%)	#Paras	Complexity*	Accuracy(%)	#Paras	Complexity*	Accuracy(%)
CSANN	785	/	83.11 ± 0.89	785	/	79.67 ± 0.83	785	/	83.22 ± 1.28
QSANN	**49**	48/48	84.79 ± 1.29	**49**	48/48	80.29 ± 1.78	**61**	48/96	84.25 ± 1.75
Ours	78	**39/39**	$\mathbf{85.33 \pm 1.43}$	78	**39/39**	$\mathbf{82.11 \pm 1.33}$	78	**39/39**	$\mathbf{84.78 \pm 0.88}$

* The number of qubits and two-qubit quantum gates. The best experimental results are highlighted in bold

the quantum-classical hybrid models ICWE-QNN [10] and QITSA [11]. Such hybrid models typically rely on core principles of quantum information theory, including superposition, entanglement, and interference, but simulate quantum behavior through classical algorithms rather than implementing explicit quantum circuits.

As reported in Table 6, on the MPQA dataset, our model achieves higher accuracy than ICWE-QNN, although it remains slightly below QITSA, while obtaining the highest F1-score of 77.10. On the SUBJ dataset, although the accuracy is relatively lower, the model still achieves the best F1-score, indicating a more balanced overall performance.

Due to the current limitations of quantum hardware, QNLP remains at an early stage of development, making direct numerical comparisons with state-of-the-art classical models inappropriate [25]. For reference, fine-tuned classical models such as BERT

Table 6 Numerical experiments performed on MPQA and SUBJ datasets

Method	MPQA		SUBJ	
	Acc(%)	F1	Acc(%)	F1
ICWE-QNN	86.80	76.00	92.60	90.90
QITSA	**87.50**	75.81	**92.96**	91.04
QSALSTM	87.15	**77.10**	91.20	**91.34**

The best experimental results are highlighted in bold

and RoBERTa typically achieve accuracies of 93–96% on SUBJ and 88–91% on MPQA. However, given the fundamental differences in model capacity and computational paradigm, direct comparison with such models is neither the objective of this work nor methodologically appropriate at the present stage.

These results nevertheless provide a useful point of reference: with only 78 trainable parameters and a compact quantum circuit design, AQLSTM achieves 91.20% accuracy on SUBJ and 87.15% on MPQA, suggesting that quantum models are approaching practically meaningful performance ranges. As quantum hardware continues to mature, the representational efficiency of quantum systems and the exponential dimensionality of Hilbert space may enable competitive performance with substantially fewer parameters.

4.3 Robustness against quantum noise

In realistic quantum computing environments, quantum noise is unavoidable, and its impact on quantum machine learning algorithms has been systematically studied in prior work, including classification tasks under noisy channels in extreme environments [26]. To evaluate the robustness of AQLSTM, we consider several representative types of quantum noise, as described as follows:

Bit-flip noise simulates the process in which a qubit flips from $|0\rangle$ to $|1\rangle$ or from $|1\rangle$ to $|0\rangle$:

$$\rho' = (1 - p_1)\rho + pX\rho X^\dagger, X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \tag{27}$$

where p_1 is the probability of bit flipping.

Depolarization noise describes the process in which the qubit state is randomly perturbed:

$$\begin{aligned} \rho' &= (1 - p_2)\rho + \frac{p}{3}(X\rho X^\dagger + Y\rho Y^\dagger + Z\rho Z^\dagger), \\ Y &= \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \end{aligned} \tag{28}$$

where p_2 is the probability of depolarization.

Table 7 Comparisons of quantum noise channel performance

Noise channel	Noise level	Accuracy(%)	
		MC	RP
Zero noise	0	100.0	80.65
Bit flip	0.01	100.0	80.65
	0.1	100.0	80.65
	0.2	100.0	74.19
Depolarization	0.01	100.0	80.65
	0.1	100.0	80.65
	0.2	100.0	74.19
Amplitude damping	0.01	100.0	80.65
	0.1	100.0	80.65
	0.2	100.0	80.65
Combinations	0.01	100.0	80.65
	0.1	100.0	80.65
	0.2	100.0	80.65

Amplitude damping noise models the energy dissipation process of qubits as follows:

$$\rho' = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix} \rho \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix} + \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix} \rho \begin{pmatrix} 0 & 0 \\ \sqrt{\gamma} & 0 \end{pmatrix}, \tag{29}$$

where γ is the probability of amplitude damping.

We set the noise intensities of the three channels to 0.01, 0.1, and 0.2, respectively, and report the corresponding results in Table 7. The results show that when the noise intensity is no greater than 0.1, the test accuracy on both the MC and RP datasets remains unchanged. When the noise intensity increases to 0.2, the test accuracy on the MC dataset remains high, whereas the accuracy on the RP dataset decreases under bit-flip and depolarization noise, while still remaining stable under amplitude damping noise.

In practical quantum hardware, multiple types of noise may occur simultaneously. To simulate a more realistic quantum computing environment, we further combine the three noise channels to evaluate their joint effect on model performance. The results show that, even under different levels of combined quantum noise, the final accuracy on the MC and RP datasets remains high, demonstrating that AQLSTM is robust against quantum noise.

4.4 Ablation study

To investigate the contribution of the proposed SP-QSA mechanism to the overall classification performance, we consider a variant without the self-attention mechanism,

Table 8 Accuracy comparisons on ablation experiments

Dataset	Index	N-QLSTM	QSALSTM
MC	Acc(%)	100.0	**100.0**
RP	Acc(%)	74.19	**80.65**
Yelp	Acc(%)	83.00	**87.50**
IMDb	Acc(%)	80.00	**84.00**
Amazon	Acc(%)	82.00	**86.50**
MPQA	Acc(%)	84.73	**87.15**
	F1 score	74.56	**77.10**
SUBJ	Acc(%)	88.40	**91.20**
	F1 score	88.40	**91.34**

The best experimental results are highlighted in bold

denoted as N-QLSTM, and conduct comparative experiments on multiple datasets using the same parameter settings and data partitions.

As shown in Table 8, AQLSTM achieves higher test accuracy than N-QLSTM across datasets of different scales. In particular, the F1-scores on the MPQA and SUBJ datasets are improved by 2.54% and 2.94%, respectively.

These ablation results indicate that the proposed SP-QSA mechanism enables the model to focus more effectively on informative words within the input sequence, thereby enhancing feature extraction and improving its ability to capture long-range dependencies and semantic associations.

5 Conclusion

In this study, we proposed, for the first time, an all-quantum long short-term memory model (AQLSTM) for text sentiment classification. By integrating a subspace projective quantum self-attention (SP-QSA) mechanism while preserving quantum state integrity, the proposed model elevates sentiment classification performance. We employ quantum embedding to map classical data into quantum states for more efficient representation and processing. Different quantum gates are then used to simulate and differentiate the queries, keys, and values in the attention mechanism through quantum-state evolution, thereby improving the model's ability to capture long-range dependencies and complex semantic associations. Experimental results across datasets of various sizes demonstrate that the proposed model achieves higher classification accuracy, reduced circuit complexity, improved scalability on larger datasets, and strong robustness in text sentiment classification tasks.

The parameterized quantum circuit architecture of AQLSTM is well aligned with the algorithmic framework of the NISQ era. Future research will focus on three main directions. First, we will optimize the quantum circuit structure to further boost computational efficiency and facilitate implementation on NISQ devices. Second, we will explore the integration of quantum image processing techniques to extend the proposed model to multimodal sentiment analysis, particularly for text-image paired sentiment

classification tasks [27]. Third, to further enhance accuracy and applicability, we plan to combine our approach with other quantum neural architectures, such as quantum convolutional networks, while expanding it to larger-scale text sentiment analysis scenarios.

Appendix A: Monotonic relationship between HSD and cosine similarity

A.1 Fundamental definitions and assumptions

Let $|\psi_q\rangle$ and $|\psi_k\rangle$ be normalized pure states with $\langle\psi_q|\psi_q\rangle = \langle\psi_k|\psi_k\rangle = 1$. Their classical counterparts $q, k \in \mathbb{R}^d$ satisfy $\|q\| = \|k\| = 1$. The quantum-classical correspondence is defined as follows:

$$|\psi_q\rangle \leftrightarrow q, \quad |\psi_k\rangle \leftrightarrow k.$$

The quantum inner product equals the classical cosine similarity, namely:

$$\langle\psi_q|\psi_k\rangle = q \cdot k = S_c \quad \text{(real-valued)}.$$

A.2 Quantum Expression of HSD

The expectation value $\langle Z_a\rangle$ from SWAP-test measurements satisfies:

$$\langle Z_a\rangle = |\langle\psi_q|\psi_k\rangle|^2 = S_c^2.$$

The Hilbert-Schmidt Distance (HSD) is defined as:

$$D_{HS} = \sqrt{2\,(1 - \langle Z_a\rangle)} = \sqrt{2(1 - S_c^2)}.$$

A.3 Monotonicity proof

D_{HS} and S_c exhibit a strictly monotonically decreasing relationship over $S_c \in [0, 1]$. We rewrite Eq. (1) as:

$$f(S_c) = \sqrt{2(1 - S_c^2)},$$

and compute its derivative as:

$$\begin{aligned} f'(S_c) &= \frac{d}{dS_c}\left[\sqrt{2(1 - S_c^2)}\right] \\ &= \frac{-2S_c}{\sqrt{2(1 - S_c^2)}}. \end{aligned}$$

For $S_c > 0$, $f'(S_c) < 0$ implies strict monotonicity. The boundary values are obtained as follows:

$$f(0) = \sqrt{2}, \quad f(1) = 0.$$

A.4 Compatibility with classical attention

Classical attention weights use softmax on cosine similarity, that is:

$$\alpha_{qk}^{\text{classic}} = \frac{\exp(S_c)}{\sum \exp(S_c)}.$$

Quantum attention weights embed HSD in exponents, formulated as:

$$\alpha_{qk}^{\text{quantum}} = \frac{\exp(-\gamma D_{HS})}{\sum \exp(-\gamma D_{HS})}.$$

Key equivalence: Since D_{HS} is monotonic to S_c, there exists an invertible mapping $g(\cdot)$ such that:

$$-\gamma D_{HS} = g(S_c).$$

Adjusting temperature parameter $\gamma > 0$ enables:

$$\exp(-\gamma D_{HS}) \propto \exp(\beta S_c), \quad \beta \propto \gamma.$$

A.5 Physical interpretation and parameters

- Temperature γ: It controls sharpness of attention distribution.
  - $\gamma \to 0$: Uniform weights (ignores similarity)
  - $\gamma \to +\infty$: Hard attention (only most similar item)
- Quantum advantage: HSD via SWAP-test achieves $\mathcal{O}(1)$ circuit depth vs. classical $\mathcal{O}(d)$ inner product.

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Declarations

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